S V V SAI SRI VALLABH PATANENI

Control Strategy Engineer · Global Graduate Program 2026

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PROFILE

Automotive Engineering Master's student at RWTH Aachen specialising in control and operating-strategy engineering for energy systems. I develop, validate and optimise control strategies in MATLAB/Simulink and C++: rule-based and predictive power-split strategies at Robert Bosch GmbH, real-time energy-management and thermal control at Ecogenium e.V., and a controller for an aircraft fuel cell system at DLR implemented for direct deployment. I am strong in model validation against measured data, multi-objective optimisation and benchmarking, and I am applying to join Goldwind's Global Graduate Program 2026 as a Control Strategy Engineer in Frankfurt.

PROFESSIONAL EXPERIENCE

Operating Strategy / System Simulation Intern
09/2025 – 12/2025
Robert Bosch GmbH — Schwieberdingen, Germany

- **Control Strategy Development:** Engineered a new MATLAB control strategy for Strong Fuel Cell Hybrid EVs that manages dynamic loads, thermal stress, low-voltage operation and SOC swing, parameterised via MAPs for different operating conditions and duty cycles.
- **Model-Based Development & Validation:** Developed and validated an iterative rule-based power-split strategy in the Simulink (FC)EVsim module, correlating simulation outputs against reference measurement data to ensure high model fidelity.
- **Optimisation & Benchmarking:** Executed multi-objective, multi-variable parametric optimisation using Genetic Algorithms and Gradient Descent to generate control MAPs, benchmarked against Dynamic Programming and ECMS standards.
- **Predictive Control & Publication:** Researching predictive control extensions using live GPS and road-elevation data to anticipate power demand; co-authoring a peer-reviewed paper on the control strategy.

Werkstudent, Fuel Cell System Control (Aircraft)
01/2026 – Present
DLR, Innovation Center for Small Aircraft Technologies (INK) — Wuerselen (Aachen), Germany

- **Control-Oriented Modelling:** Built a MATLAB/Simulink digital twin of the aircraft fuel cell thermal management system to derive control specifications (volumetric flow, coolant temperature range, coolant pressure) for the ~150 kW system.
- **Controller Development (Thesis):** Developing the fuel cell system control (start-up, purge control, cathode and anode subsystem control, temperature control, safety protocols) in MATLAB and C++ for direct deployment to the physical controller.
- **System Integration:** Selecting subsystem components against derived specifications and designing the system rig in CATIA V5 for integration and test.

Control & Energy-Management Lead (Team Lead)
06/2024 – Present
Ecogenium e.V (RWTH student team), Shell Eco-Marathon — Aachen, Germany

- **Real-Time Control Strategy:** Architected and deployed energy-management and fuel-cell temperature-control strategies in Python/Simulink, validated with Model-in-the-Loop simulation, boosting on-track efficiency by ~80% and securing a runner-up finish at Shell Eco-Marathon 2025.
- **Control Validation on Hardware:** Engineered and programmed a standalone test bench and correlated control models against real measurement data (polarisation curves, purging, thermal sweeps) to validate strategy before deployment.
- **Leadership:** Led a multidisciplinary team building Germany's only student-built hydrogen fuel cell vehicle from concept to competition, coordinating control, integration and on-track troubleshooting under deadlines.

Autonomous Systems Engineer (Academic Project)
04/2025 – 07/2025
Institute for Automotive Engineering (IKA), RWTH Aachen — Aachen, Germany

- **Controller Design & Validation:** Developed models and tuned PID and MPC-based controllers in MATLAB/Simulink and Python for Adaptive Cruise Control and Lane Keep Assist, validated through real-time testing on a mock vehicle and in CARLA against real-world results.

KEY SKILLS

Control & Strategy	control strategy design · rule-based & predictive control · PID / MPC · operating strategy · energy management (EMS) · real-time control · MIL validation
Optimisation & Data	multi-objective optimisation · Genetic Algorithms · Gradient Descent · DP / ECMS benchmarking · MAP / lookup-table calibration · KPI analysis
Simulation & Modelling	MATLAB/Simulink · digital twin · model validation · Dymola · CARLA
Programming	MATLAB · C/C++ · Python (NumPy, Pandas, scikit-learn, PyTorch)
Data & Tools	Power BI · Tableau · Siemens Teamcenter · Git

EDUCATION

M.Sc. Automotive Engineering
09/2023 – 06/2026
RWTH Aachen University — GPA 1.7 (1.2 core courses)

Modules: Simulation Sciences, Mechatronics, Machine Learning / Data Science, Alternative & Mobile Propulsion Systems. Thesis: fuel cell system control (MATLAB/C++).

B.Tech Mechanical Engineering
06/2019 – 04/2023
Jawaharlal Nehru Technological University, Hyderabad — GPA 1.3 (German scale)

SELECTED PROJECTS & PUBLICATIONS

- Co-author (in preparation): rule-based powertrain control strategy with optimisation and benchmarking, based on the Bosch internship.
- Vehicle Sensor Data Analysis (IKA-RWTH): Python pipelines for camera/LiDAR/radar, CNN traffic-sign recognition, Power BI dashboards.
- Paper: A Comprehensive Analysis of Two Innovative Aircraft Design Configurations (IJSED).

LANGUAGES

English (C1, fluent) · German (B1, intermediate) · Telugu / Hindi (native)